ENGINEERING TECHNICAL PROJECT REPORT

Engineering Assessment of Land Use Transformation and Green Space Degradation Using GIS RUET Campus (2003–2023)

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Project Type:

Technical Project Report (GIS & Remote Sensing Analysis)

Tools & Software Used:

ArcGIS | Remote Sensing (Landsat Data) | Spatial Analysis Techniques

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Abstract:
Rapid infrastructural development within institutional environments often leads to significant environmental challenges, particularly the reduction of green spaces. This study presents a comprehensive analysis of temporal land-use and land-cover (LULC) changes within the Rajshahi University of Engineering & Technology (RUET) campus over 20 years (2003–2023). Using Remote Sensing (RS) and Geographic Information System (GIS) techniques, the study quantifies the changes in vegetation, built-up areas, and water bodies.
The results indicate a 10.64% decline in green space, primarily due to infrastructural expansion. Additionally, future plantation strategies (25%, 50%, and 75%) are evaluated to assess their effectiveness in restoring greenery. The findings highlight the need to integrate sustainable planning practices into campus development.

1. Introduction:

Urban green spaces play a crucial role in mitigating environmental degradation by reducing urban heat island effects, improving air quality, and maintaining ecological balance. However, rapid infrastructural expansion often results in a significant decline in vegetative cover, particularly in institutional campuses undergoing continuous development.
The RUET campus has experienced substantial infrastructural growth over the past two decades, including the construction of multiple high-rise buildings. While this development supports academic expansion, it has raised concerns regarding the reduction of green space and its impact on the campus environment.
This study employs GIS and Remote Sensing techniques to evaluate temporal changes in Land Use Land Cover (LULC) and assess the environmental implications of such transformations. Furthermore, it proposes sustainable strategies to restore and maintain green space within the campus.

2. Objectives:

The primary objectives of this study are:

- To analyze temporal LULC changes using satellite imagery
- To quantify variations in green space, built-up area, and water bodies
- To assess the environmental impacts caused by urban expansion
- To evaluate green space recovery potential through plantation strategies
- To support sustainable campus planning and decision-making

3. Study Area:

The study area is the RUET campus located in Rajshahi, Bangladesh. The geographical extent is approximately:

- Latitude: 24°21′45″N to 24°22′30″N
- Longitude: 88°37′30″E to 88°38′00″E

The campus has undergone continuous infrastructural development, making it suitable for analyzing land cover dynamics and environmental changes.

4. Data Sources and Tools:

- **Satellite Data:** Landsat imagery (2003, 2013, 2023)
- **Source:** USGS Earth Explorer
- **Software:** ArcGIS
- **Techniques Used:**
  - Supervised Classification
  - Maximum Likelihood Algorithm
  - Raster-to-Vector Conversion
  - Spatial Analysis

5. Methodology:

5.1 Workflow Overview:

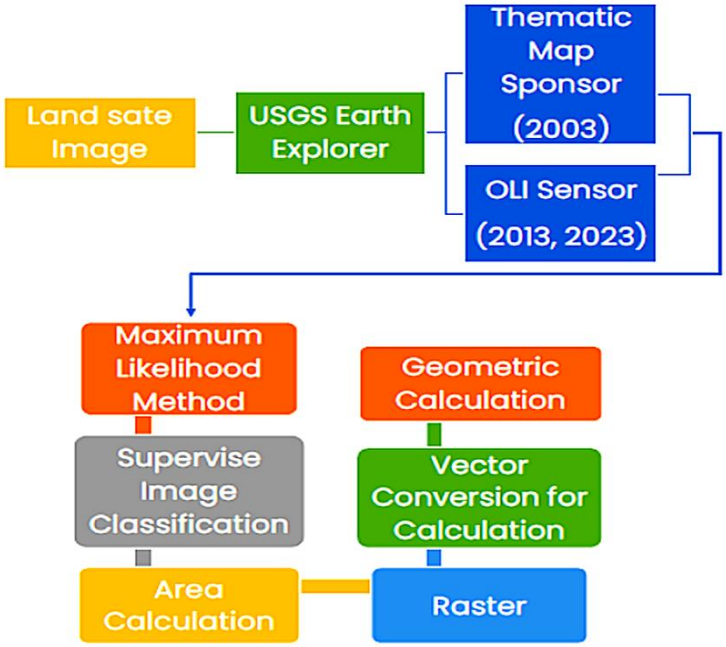

Figure 1: *Workflow of Remote Sensing-based LULC analysis.*

The methodological framework adopted in this study consists of the following sequential steps:

- Satellite image acquisition
- Image preprocessing
- Supervised classification using the Maximum Likelihood Method
- Raster-to-vector conversion
- Area calculation
- Comparative temporal analysis

5.2 Image Classification:

A supervised classification approach was implemented using the Maximum Likelihood Method, which classifies pixels based on statistical probability distribution. This method was selected due to its robustness and widespread application in medium-resolution satellite image classification.

The land cover classes considered in this study include:

- Vegetation (Green Space)
- Built-up Area
- Water Bodies

5.3 Area Calculation:

After classification:

- Raster images were converted into vector format
- Area calculations were performed using GIS tools
- Comparative analysis was conducted for the years 2003, 2013, and 2023

Following the methodological framework, the results obtained from the LULC classification are presented in the subsequent section.

6. Results and Analysis:

6.1 Land Use Land Cover Maps:

This section presents the spatiotemporal analysis of Land Use and Land Cover (LULC) changes within the RUET campus from 2003 to 2023 using GIS and remote sensing techniques. The results are interpreted through thematic maps, quantitative comparisons, and graphical representations to evaluate variations in vegetation, built-up areas, and water bodies.

A comparative assessment of Figures 2–4 reveals a progressive transformation in land cover distribution over the study period.

As illustrated in Figure 2, the 2003 condition is dominated by vegetation, which accounts for 68.84% of the total area, while built-up areas occupy 23.79% and water bodies represent

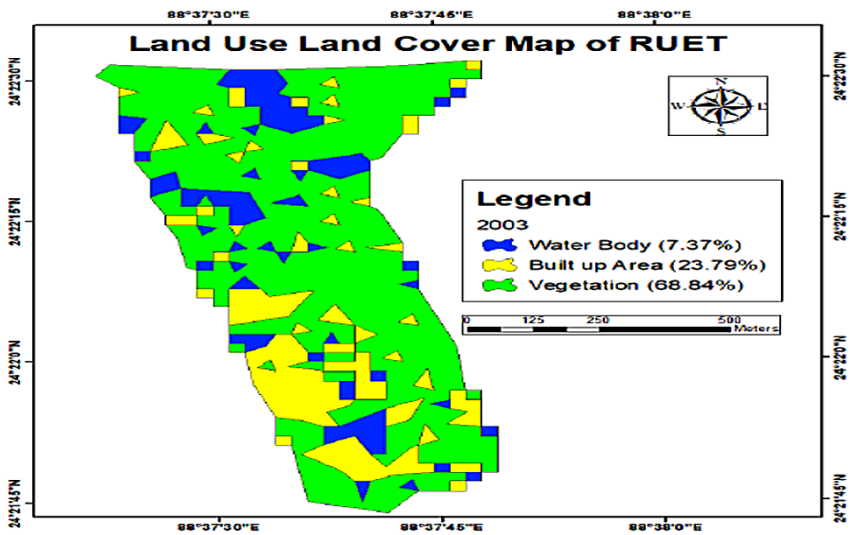

Figure 2: *LULC map of RUET campus (2003)*

7.37%. This distribution indicates that the campus initially maintained a relatively balanced environmental state, with vegetation acting as the primary surface component.

A noticeable transition is observed in Figure 3, where built-up area increases to 31.15%, reflecting the onset of significant infrastructural expansion. Although vegetation decreases only marginally to 68.25%, the spatial redistribution suggests that development has begun to encroach upon previously vegetated zones. The sharp decline in water bodies to 0.60% indicates possible land modification, infilling, or seasonal variability in satellite imagery.

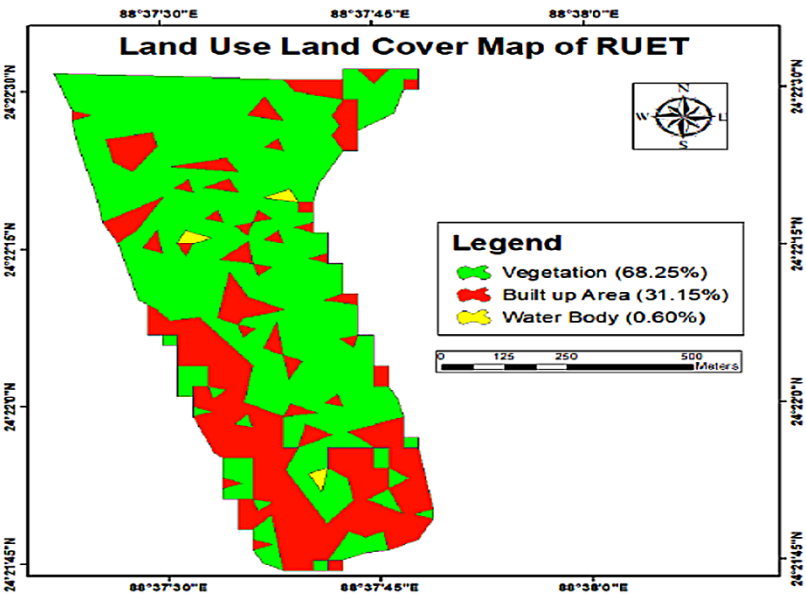

***Figure 3**: LULC map of RUET campus (2013)*

By 2023, as presented in Figure 4, the transformation becomes more pronounced. Vegetation coverage declines to 58.20%, while built-up area increases further to 35.80%, confirming a continuous shift toward an infrastructure-dominated landscape. Water bodies show partial recovery to 5.99%, which may be attributed to localized hydrological conditions or temporal variations.

The progression observed across Figures 2–4 demonstrates a consistent conversion of vegetated surfaces into built-up areas, indicating that infrastructural development is the dominant driver of land cover change within the campus.

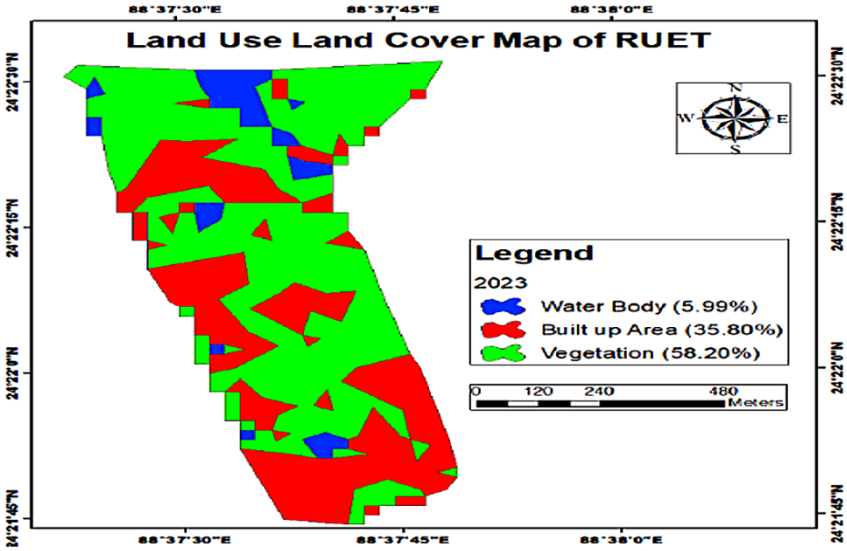

Figure 4: *LULC map of RUET campus (2023)*

6.2 Temporal Change Analysis:

The temporal variation of land cover classes, summarized in Figure 5, highlights a clear long-term trend characterized by a steady decline in vegetation and a corresponding increase in built-up area.

Between 2003 and 2023, vegetation cover decreased by approximately 10.64 percentage points, while built-up area increased significantly over the same period. This trend indicates a gradual replacement of permeable and environmentally active surfaces with impervious construction-related surfaces.

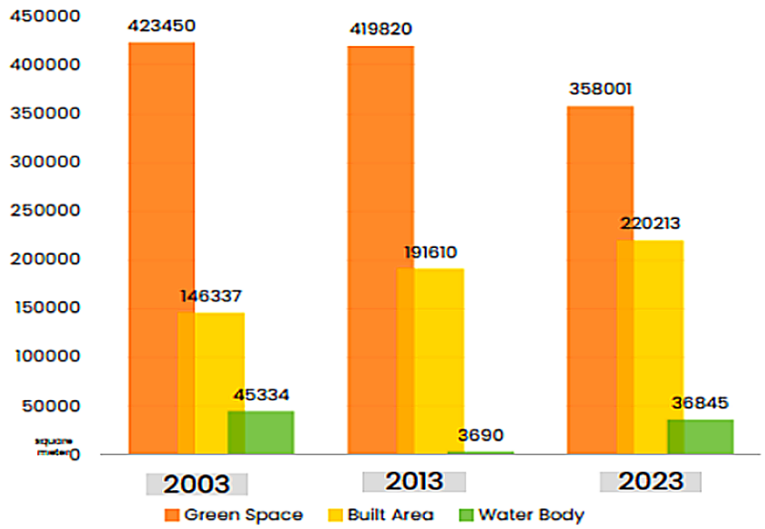

Figure 5: *Temporal variation of land cover (2003–2023)*

The pattern illustrated in Figure 5 suggests that the rate of land transformation intensified during the later stages of development, particularly between 2013 and 2023. This implies that recent infrastructural expansion has occurred at a higher density or over larger spatial extents compared to earlier phases.

Water bodies exhibit irregular fluctuations in Figure 5, indicating sensitivity to seasonal variation, drainage conditions, and possible land-use alterations. Unlike vegetation and built-up areas, their behavior does not follow a strictly linear trend.
Overall, the temporal analysis confirms that the campus is undergoing a systematic shift toward increased surface impermeability, which has direct implications for environmental performance.

6.3 Area-Based Analysis:
The area distribution presented in Figure 6 provides a more detailed understanding of the structural nature of land transformation within the campus.

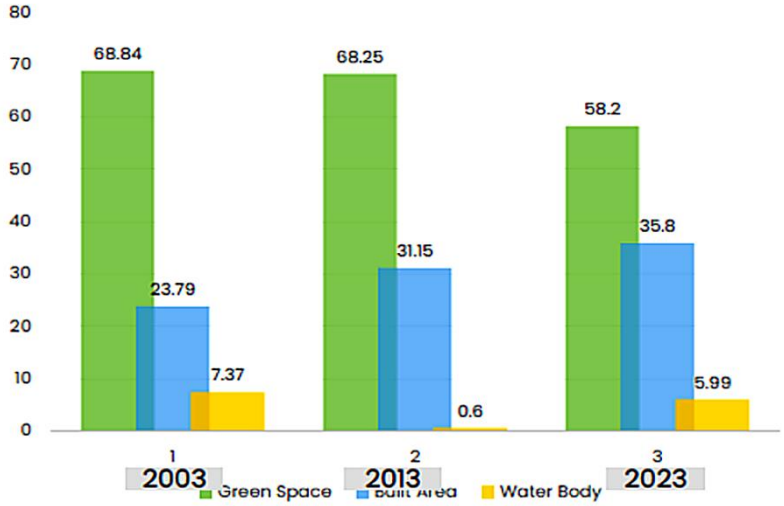

Figure 6: *Land cover area distribution (Sq.m)*

The results demonstrate a strong inverse relationship between vegetation and built-up areas, indicating that the expansion of infrastructure is directly associated with the reduction of green space. This relationship reflects a conversion-driven process, where previously vegetated land is being replaced by built-up surfaces rather than undergoing independent variation.
From an engineering perspective, this transformation alters the functional characteristics of the surface system. Vegetation contributes to environmental regulation through evapotranspiration, shading, and infiltration, whereas built-up surfaces are associated with increased heat absorption, reduced permeability, and higher runoff generation.
The trends observed in Figure 6 therefore indicate not only a change in land cover distribution but also a shift in the environmental performance of the campus, moving toward a less sustainable surface configuration.

6.4 Green Space Recovery Potential:

The impact of different plantation strategies is evaluated in Figure 7, which compares existing green coverage with projected improvements under varying intervention scenarios.

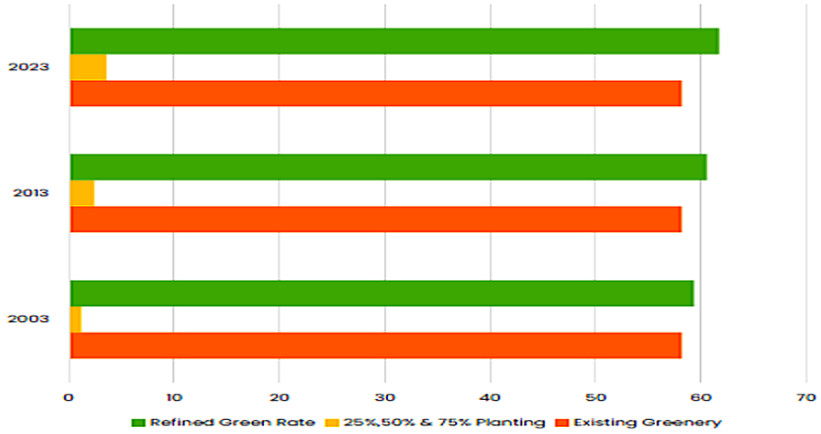

***Figure** 7: Existing vs refined green rate under plantation scenarios*

The results indicate that increased plantation intensity leads to measurable improvement in vegetation coverage. Among the evaluated scenarios, the 75% plantation strategy produces the most significant outcome, increasing green space by approximately 3.58% and recovering nearly 33.46% of the lost vegetation.

Figure 7 demonstrates that although complete restoration of the original land cover condition is not feasible under current spatial constraints, substantial recovery can be achieved through targeted intervention. The results highlight the effectiveness of strategic green infrastructure planning in mitigating the environmental impact of infrastructural expansion.

7. Discussion:

The results presented in Figures 2–7 collectively demonstrate a systematic transformation of the RUET campus driven by continuous infrastructural development.

The spatial patterns observed in Figures 2–4 indicate that built-up areas have progressively expanded over previously vegetated land, leading to a reduction in green space. This trend is quantitatively confirmed in Figure 5, where vegetation shows a consistent decline and built-up area increases steadily over the study period.

The inverse relationship between vegetation and built-up areas, clearly illustrated in Figure 6, indicates that land conversion is the primary mechanism of change. This suggests that the expansion of infrastructure has occurred directly at the expense of ecological space, rather than through independent fluctuations in land cover classes.

The variation in water bodies observed between Figures 3 and 4 suggests that these features are influenced by localized environmental conditions and seasonal variability, and therefore do not follow the same systematic trend as vegetation and built-up areas.

From an engineering standpoint, the increasing dominance of impervious surfaces, as indicated by Figures 5 and 6, has significant implications. Reduced infiltration capacity may lead to increased surface runoff and higher demand on drainage systems, while the decline in vegetation reduces natural cooling and environmental buffering capacity. These changes indicate a transition toward a higher environmental load condition within the campus.
The plantation scenarios presented in Figure 7 provide a practical framework for mitigating these effects. The results demonstrate that strategic green interventions can significantly improve vegetation coverage and partially restore ecological balance, even within the constraints of ongoing development.

8. Sustainable Planning Recommendations:

8.1 Green Infrastructure Development:

- Implement a 75% plantation strategy in available spaces
- Expand vegetation in open and roadside areas

8.2 Rooftop and Vertical Greening:

- Introduce rooftop gardens
- Develop vertical green walls

8.3 Structural Considerations:

- Use lightweight planting systems
- Prefer low-load perennial plants

8.4 Strategic Planning:

- Focus planting in heat-prone zones (e.g., west-facing buildings)
- Encourage dense tree plantation in open areas

9. Conclusion:

This study demonstrates the effectiveness of GIS and remote sensing techniques in evaluating land cover transformation and its implications for environmental sustainability. The results, supported by Figures 2–7, confirm that infrastructural expansion within the RUET campus has led to a significant reduction in vegetation and an increase in built-up area. This transformation reflects a shift toward increased surface impermeability and reduced environmental performance. However, the scenario analysis presented in Figure 7 indicates that strategic plantation and green infrastructure planning can significantly improve vegetation coverage and partially restore ecological balance.The study provides a data-driven framework for integrating environmental considerations into campus development and highlights the importance of sustainable planning in managing future infrastructure expansion.

10. Skills Demonstrated:

- GIS Spatial Analysis (ArcGIS)
- Remote Sensing & Satellite Data Processing

- Supervised Image Classification
- LULC Mapping
- Environmental Impact Assessment
- Data Visualization

11. Limitations of the Study:

- Classification accuracy depends on satellite image resolution
- Seasonal variation may affect water body detection
- Limited ground truth validation
- Small-scale features may not be fully captured